StepBro / Layer 1

Contents

1. Connect and make your first pattern
2. The panel and the two grid modes
3. StepBro and Yummy Gummy Synthesizer: two levels
4. Edit hits velocity swing and note length
5. Record notes and rolls
6. One bar or two bars
7. Select copy and erase pattern slots
8. Chain patterns
9. Perform with density and the loop window
10. Mute and erase sounds
11. MIDI channels transport and timing
12. Saving lights and troubleshooting
13. Firmware updates

1. Connect and make your first pattern

Layer 1 sends MIDI to instruments in your computer. Connect the device by USB, route StepBro's MIDI (Midi Channel 1) to a drum kit or synthesizer, and send MIDI clock and transport from your music software to the device. StepBro / Layer 1 comes with the Yummy Gummy Synthesizer on Layer 2, which can be accessed by pressing SHIFT + BANK + PLAY. Both Layers run at the same time. Layer 2 has its own manual, please check it for more details.
The same name is used for the MIDI port and for Yummy's audio input, whichever level the panel shows. Select the port that matches your board.

1. Connect a USB data cable. In your music software, enable MIDI input from Step Mother and enable MIDI clock or Sync output to the same device.
2. Create an instrument track receiving on Midi Channel 1. Choose a drum kit or synthesizer and enable its input monitoring.

3. Tap Note, then tap a pad. It should play a sound and select that sound for step editing.
4. Tap Pattern briefly. Tap pads 1, 5, 9 and 13 to place hits. Tap an active step again to remove it.
5. Start your software's transport. StepBro follows its clock. Use StepBro's Play key to stop StepBro while the software keeps running; press Play again and StepBro comes back in on the next beat.
6. Stop the software, then allow a few seconds for saving before unplugging.

StepBro has no internal tempo. Its sequencer needs incoming MIDI clock and a running host transport. You can audition notes while stopped.

2. The panel and the two grid modes

Key	Use
Note	Tap for sounds; double-tap to arm recording. Hold for the sound overview or as a modifier.
Pattern	Tap for steps; hold for slots; double-tap to chain multiple patterns.
Shift	Modifier for rolls, swing, the loop window, copying, mutes and the MIDI channel.
Mute	Mute sounds; with Shift, mute StepBro (pad 1), Yummy Midi Out (pad 2) or Yummy Audio (pad 3). Hold with the knob for note length.
Bank	Switch the visible half of the sounds or steps.
Clear	Erase with another key or pad. A plain tap does not erase.
Play	Stop StepBro, or bring it back in on the next beat, while the host runs; hold with the knob while the host is stopped for MIDI Lead (MIDI delay adjustment).
Knob	On its own: in Note mode, the selected sound's velocity; in Pattern mode, density. With a key or pad held: the value that combination sets.

Pads are numbered 1–16, left to right and top to bottom. "Hold X + tap Y" means keep X down while tapping Y. The Yummy manual names four of the same keys differently: Shift is Glide, Mute is Accent, Bank is Wave and Clear is Function. There are semantic overlaps between the names so it should become fairly intuitive which button does what.



Note

Note mode — pads are sounds

Each pad selects and plays one sound. Bank switches between sounds 1–16 and 17–32. Hold Note to see which sounds contain programmed hits.



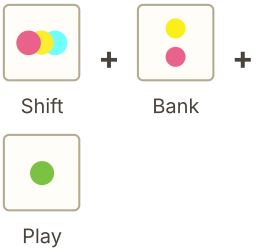
Pattern mode — pads are steps

The grid shows 16 steps for the selected sound. Bank switches between steps 1–16 and 17–32. Each step is a sixteenth note.

A quick Pattern tap from Note mode opens step editing. A longer hold opens slots after about 0.15 seconds and returns to Note mode when released.

3. StepBro and Yummy Gummy Synthesizer: two levels

The panel shows one of two levels at a time: StepBro, or the Yummy Gummy Synthesizer. Both keep playing whichever level you see; switching only changes what the keys and pads control.



Switch between StepBro and Yummy

Hold all three together, in any order. Release them all before playing on. The Bank LED blinks slowly while the Yummy level is shown. The switch never starts, stops or mutes either engine.

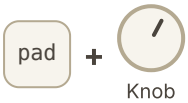
StepBro has 16 pattern slots, 32 sounds per pattern, on one MIDI channel (channel 1 at first). Power-up shows the StepBro level. Yummy plays its own sound through the board's USB audio and uses MIDI channel 2. MIDI channels can be changed for each of the Layers (see the corresponding section below).
From the StepBro level, Shift + Mute also mutes Yummy's MIDI and audio; see Mute and erase sounds.

4. Edit hits velocity swing and note length



Place or remove a hit

In Pattern mode, tap a pad. The toggle happens on release. A new hit uses velocity 100. Holding without moving the knob still toggles on release; use hold-and-turn to edit velocity.



Set step velocity

Hold a step in Pattern mode and turn the knob: 1–127. An empty step is created at that velocity. Releasing after turning leaves the hit on. You can hold several steps to adjust them together.



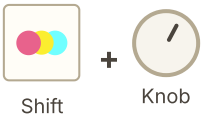
Set a sound’s velocity, programmed hits included

In Note mode, with nothing held, turn the knob. If the selected sound has hits, the knob position is its loudest hit and the other hits follow in proportion: soft and accented hits keep their difference, the whole range up to 127 is reachable and nothing clips. Auditioning and live recording use the same velocity. Without hits, the knob sets just that velocity. The knob picks up: nothing changes until it passes the current value (the loudest hit), so a knob left elsewhere never jumps it. Turning back to where you started restores the hits as they were. The selected pad’s brightness shows the velocity.



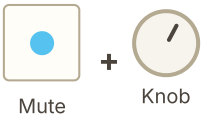
Set live playing velocity

In Note mode, hold the sound pad and turn the knob. Note does not need to remain held. This sets that sound’s audition and live-record velocity, not its already programmed steps. Turning is silent; press the pad again to hear the result.



Set swing

Hold Shift with no pads held and turn the knob. The current pattern ranges from 50% straight timing to 75% shuffle. Swing is stored with each pattern.



Set note length

Hold Mute with no pads held and turn the knob. Note length runs from 5% to 100% of a step (default 80%) and is stored with each pattern; this matters most when StepBro plays a synth line. It changes only once the knob moves, so pressing Mute alone never alters it. When no tempo has been measured yet, auditioned notes last 150 ms.

Knob controls wait until you move the knob after holding the key or pad. They then use its absolute position. Two exceptions: the knob on its own in Note mode picks up at the current velocity, and the knob during a roll moves on from the current velocity. Sound shaping belongs to the receiving instrument.

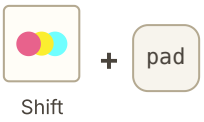
5. Record notes and rolls



Double-tap to arm recording

Make two clean, quick taps of Note. The Note LED blinks while armed. With the host running and StepBro locally playing, note pads are recorded at the nearest step, and a playhead runs across the dark pads, brighter on pad 1 at the start of each bar, so you can see when to come in. Existing hits elsewhere remain unchanged.

Tap Note once to disarm. Entering Pattern mode also disarms. Using Note as a held modifier does not count as a recording tap.



Play a sixteenth-note roll

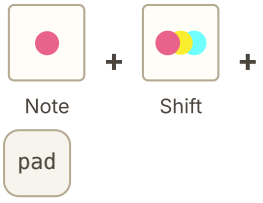
In Note mode, hold Shift, then a sound pad. The first hit sounds immediately; subsequent hits follow straight sixteenth notes while playing, even if the pattern has swing. Release the pad to stop. Multiple pads can roll together.

With recording armed, the roll is recorded. Without recording armed, Shift + pad is a performance roll and writes nothing.



Ride the roll's velocity

While a pad rolls, turn the knob to make the repeats softer or harder. It moves on from the current velocity with no jump, and turning to either end reaches 1 or 127 wherever the knob started. A recorded roll keeps the velocities you played. Swing is not changed while a pad rolls.



Record a roll without arming

Hold Note and Shift, then a sound pad. While the sequencer plays, each roll hit is added to its step. From Pattern mode, releasing Note returns to step editing.

Rolls can sound while stopped, using the last measured step duration or a 125 ms fallback before any tempo is measured. Nothing is recorded while stopped.

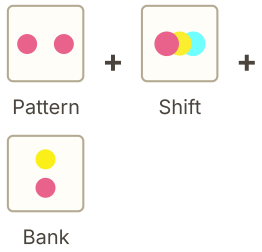
6. One bar or two bars

Each pattern stores 32 sounds over 32 steps. It plays 16 steps when the second step bank is empty, or 32 steps when any sound has a hit in steps 17–32. Merely viewing the second bank does not lengthen playback. These are one and two bars in 4/4 time.



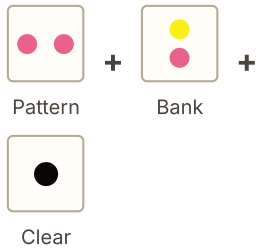
Show the other half

In Note mode: sounds 1–16 or 17–32. In Pattern mode: steps 1–16 or 17–32. A pattern with no hits in steps 17–32 plays one bar.



Copy the visible step bank

In Pattern mode, hold Pattern and Shift, then tap Bank. All 16 visible steps for every sound replace the other bank. The destination's previous hits are overwritten.

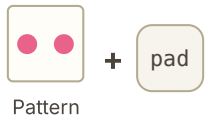


Clear the visible step bank

In Pattern mode, hold Pattern and Bank, then tap Clear. Every sound is cleared in the visible 16-step half only. Clearing all hits in steps 17–32 returns the pattern to 16-step playback. There is no undo.

The sound bank selection is saved with the pattern. The visible step bank resets to the first half on a manual slot recall.

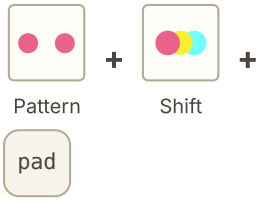
7. Select copy and erase pattern slots



Select a slot

Hold Pattern until slots appear, then tap a slot. Empty slots are dark; populated slots glow dimly; the active or queued slot has a quick repeating dip. Release Pattern when finished.

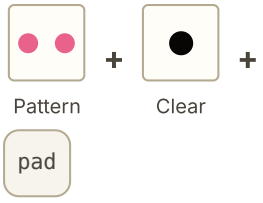
While the host plays, manual slot changes wait for the end of the loop: 16 steps, or 32 when the active pattern uses two bars. This also applies while StepBro is stopped with Play, because it keeps counting with the host. With the host stopped, selection is immediate.



Copy a slot

Hold Pattern and Shift. Tap the source, then the destination. The source pulses while waiting. Copying replaces notes, velocities, swing, note length and sound bank, and also copies current sound mutes. Release Pattern to cancel an unfinished copy.

The destination becomes active, immediately when the host is stopped or at the end of the loop while it plays. If a pattern chain exists, then copying and selecting a slot end chaining.



Erase a whole slot

Hold Pattern and Clear, then tap the target slot. All 32 sounds and all 32 steps are erased and its pattern settings reset. This is immediate and has no undo. Other slots remain intact.

8. Chain patterns


Pattern

Double-tap for chain entry

Double-tap Pattern cleanly. The Pattern LED blinks and the pads show the slots. Entering chain mode disarms note recording.

Tap slots in the order you want them to play. Repeats count as separate entries: 1, 1, 2, 3, 3, 5 makes six entries. A chain holds up to 32 entries.


Pattern

Tap again to finish

Tap Pattern, or press Note, to finish chain entry. If playing, the new chain begins at the end of the current pattern or chain entry. If stopped, it is ready from its first entry.

Each entry plays its complete 16- or 32-step pattern, then advances, and the chain loops. MIDI Start resets the chain to entry 1. MIDI Continue resumes rather than issuing that reset. While StepBro is stopped with Play, the chain keeps its place with the host.
To stop a chain, manually select a slot, or enter chain mode and leave without tapping any slot. While the host plays, chain cancellation waits for the end of the loop. **Chains are not saved across power cycles - meaning when you turn off the device the chaining is gone.**

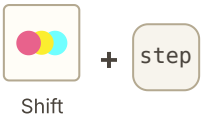
9. Perform with density and the loop window


Knob

Thin the pattern out, or fill it back in

In Pattern mode, turn the knob with nothing held, on the StepBro level. Fully right plays every hit. Turning left drops the weakest hits first, down to the strongest eighth of the pattern at the far left.

Each hit's strength is its velocity, plus extra weight on a beat (steps 1, 5, 9 and 13) and a little on an off-beat eighth. Low ghost notes and off-beat sixteenths go first; accented hits on the beat stay longest. Turning back right brings everything back.
Density only decides what plays. It never changes the stored pattern and is not saved; the board starts at full density after power-up. Muted sounds are not counted. Like the other knob controls, density waits until you move the knob, so letting go of Shift or a pad, or returning from the Yummy level, never makes it jump. In Note mode the knob on its own sets velocity instead; density stays where you left it.



Shift

Repeat one step

In Pattern mode, hold Shift and press a step. That step plays over and over, in time, for as long as you hold it.

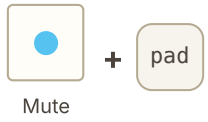


Loop a range of steps

Hold Shift, hold one step and press a second: the steps from one to the other loop. You can let go of the steps; the loop runs for as long as Shift is down.

Let go of Shift and the full pattern carries on exactly where it would have been: the loop only changes which step StepBro plays, while the host's timeline keeps counting. The loop starts on the next sixteenth and uses the 16 steps in view; press Bank first to loop steps 17–32. The looped steps glow dimly. It affects StepBro only; Yummy keeps playing. Pads used for the loop never toggle their steps, and the loop is not saved.

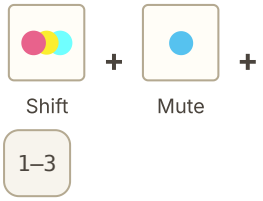
10. Mute and erase sounds



Mute a sound in the current pattern

Hold Mute and tap a sound pad in either grid mode. Pads follow the Note bank, even while editing steps. The tapped sound also becomes selected. Muted sounds flash in the Mute view.

Sound mutes belong to individual slots during the session. Muting hats in one slot does not mute them in every slot. Copies include these mutes, but mutes are not retained after power-off.



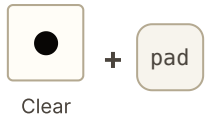
Mute StepBro or Yummy MIDI Out or Yummy Audio Out

Hold Shift and Mute, then tap a pad. Pad 1 mutes StepBro. Pad 2 mutes Yummy's MIDI notes; notes already sounding get their note-off at once, so nothing hangs. Pad 3 mutes Yummy's audio with a short fade, echoes included. Muted pads flash. Everything keeps its place underneath, so unmuting comes back in time. These mutes are not saved.



Unmute the current pattern's sounds

Hold Clear and tap Mute. This clears sound mutes in the active slot only. It does not undo Shift + Mute or unmute other slots.



Erase one sound's steps

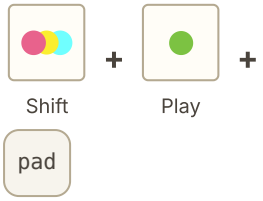
In Note mode, hold Clear and tap a sound pad. All 32 steps for that sound in the current slot are cleared. Other sounds remain.



Erase every sound in the active slot

Hold Shift, press and release Clear while Shift remains held. This removes the current slot's hits. For a full slot reset, use Pattern + Clear + slot. Neither operation has undo.

11. MIDI channels transport and timing



Choose StepBro's MIDI channel

Hold Shift first, then Play. Pads 1–16 select channels 1–16; the bright pad is the current channel. Yummy uses channel 2, so keep StepBro on another channel if your software listens to both.

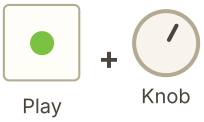
Sounds 1–32 send MIDI notes 36–67 on StepBro's channel, in ascending semitones. Bank changes the visible 16-sound window. Octave names vary by software: these are C1–G3 in Ableton's convention. Use note numbers when matching another application.



Stop StepBro, then bring it back on the beat

While the host runs, press Play to stop StepBro at once. It goes silent but keeps counting with the host, chains included. Press Play again: the Play LED blinks, and StepBro comes back in on the next beat, on the step the host is on. Press Play again while it blinks to cancel. Start from the host to reset playback; Continue resumes the position.

This source handles MIDI Clock, Start, Continue and Stop. It does not implement MIDI Song Position Pointer. Do not assume it jumps to a host song position when starting midway through a song.



Adjust MIDI lead while the host is stopped

Stop the host. Hold Play and turn the knob to set 0–16 ms in 1 ms increments. The grid lights one pad per millisecond; no pads means 0 ms, all 16 means 16 ms. Default is 4 ms (four pads lit). The setting is saved with your patterns.

Lead sends predicted sequencer notes earlier to compensate for host and USB latency. Rolls riding the step clock follow it; manually played notes are immediate. The first step after Start cannot be predicted early; a restart with Play is. Record a test and adjust lead to your setup. Play + knob does not change lead while the host transport runs.

12. Saving lights and troubleshooting

Patterns, step velocities, swing, note length, sound bank, the last active slot, the MIDI channel and MIDI lead are saved, together with all of Yummy's work, in one save. Saving happens about 1.5 seconds after the last change, once the host transport and Yummy are both stopped and no keys are held. Saving is silent: there is no light for it. Allow a few seconds after stopping before disconnecting.
Chains, sound mutes, the Shift + Mute mutes, density, the loop window and live audition velocity settings are session controls and are not saved. Programmed hits changed with the velocity knob are saved like any other edit. The panel starts on the StepBro level after power-up.

Indicator	Meaning
Note LED blinking	Recording armed.

Indicator	Meaning
Pattern LED blinking in chain entry	Pads are adding chain entries.
Bank LED blinks slowly	The Yummy level is shown.
Play LED blinks quickly while the host runs	StepBro is stopped and will come back in on the next beat.
Flashing pad 2 or 3 in the Shift + Mute view	Yummy's MIDI or audio is muted.
Quick dip in slot view	Active or queued slot.
Flashing sound in Mute view	Sound is muted in this slot.
Pulsing slot while copying or erasing	Copy source or erase target view.
Brief pad flash	A sound or chain entry was triggered or selected.
Dim pad moving across Note mode, brighter on pad 1	Recording armed: the playhead, and the start of each bar.
Brightness in step view	Hit velocity; the current step also receives playhead feedback.

Problem	Check
Pads send no sound	Route the matching USB MIDI port and channel to an instrument and enable monitoring. StepBro itself produces no audio.
Sequence will not run	Enable clock/Sync output to the board and start the host transport. If StepBro was stopped with Play, press Play and wait for the next beat.
Expected sound missing	Check the sound bank, MIDI mapping, the current slot's sound mutes, Shift + Mute pad 1, and the knob's density setting.
Loop stays one bar	Place at least one hit in steps 17–32. Viewing bank 2 alone does not extend it.
Slot change seems delayed	Manual changes wait for the end of the 16- or 32-step loop; chain entries use their own pattern length.
Chain, mutes, density or the loop window reset after restart	Expected: they are not saved.
Hits missing during playback	In Pattern mode, turn the knob fully right (full density).
Notes cut short	Hold Mute and turn the knob to the right to lengthen the pattern's note length.

Problem	Check
Pattern edits lost after restart	Stop the host, release all keys and wait a few seconds. If edits are still lost, switch to the Yummy level, record FUNCTION + PLAY into a MIDI clip and export it: the report shows why the save did not happen.
LEDs all remain dark	The firmware can still send MIDI if the LED driver is unavailable. Confirm MIDI operation and report the hardware fault.

13. Firmware updates

StepBro and Yummy are one firmware. Updating it updates both and keeps your saved work. Use StepDaddy, the updater, on macOS or Windows: open it and follow its instructions. It works out which board you have, or asks: the Step Mother 1 has a micro-USB socket, the Step Mother 2 a USB-C socket.



Install mode at power-up

Disconnect USB. Hold Shift and Clear (Glide and Function on the Yummy level) while reconnecting, and keep them down for half a second. StepDaddy asks for this when it needs it. The combination is checked only at power-up; during normal operation Shift + Clear erases the current slot's hits.

Step Mother 1 (Teensy, micro-USB): StepDaddy can usually restart the board into install mode on its own, with no keys. If it asks, use Shift + Clear at power-up.
Step Mother 2 (RP2040, USB-C): use Shift + Clear at power-up; the board appears as a drive called RPI-RP2, StepDaddy copies the firmware onto it and the board restarts. A brand-new board opens RPI-RP2 by itself.
Updates keep your patterns, settings and Yummy's work. The firmware version is shown by StepDaddy.